Quality–Alignment Decoupled Learning for AIGC Image Assessment

a Quality-aware multimodal pretraining

SafeAug 1
SafeAug 2
Clean view

Shared visual backbone
CLIP ViT-L/14 + ResNet-50
Semantic–Structure Fusion

z_1, z_2

Shared quality projector P_Q
Quality predictor D_Q
I–I symmetric SimSiam

q_c

Clean quality embedding q_c
Component anchors + ordinal
Five-level quality text CE

Separate Clean scoring paths
Quality MOS + Artifact / Aesthetics
Global / local Alignment supervision

z_c → same P_Q

Clean backbone features; not P_Q output

Prompt / Answer / region phrases
Frozen CLIP text + learnable contexts

Independent P_I / D_I and P_T / D_T; relation-specific adapters
Image–Prompt · Image–Answer · Prompt–Answer · region–word

P_Q → Quality Adapter initialization

Semantic projectors / predictors discarded

b Downstream assessment · Large CLIP · ResNet layer3 · two outputs

Image
224 × 224

CLIP visual ViT-L/14
global feature: 768-D

ResNet-50 layer3
1024 × 14 × 14
1×1 proj → 256 × 14 × 14

Semantic–Structure Fusion
global query / local K,V
z: 256-D

Shared local projection L
P_0: 196 × 256

Prompt + shared learnable contexts
Frozen CLIP text encoder → U (768→256)

U(text)

Visual / Fusion / local L shared by Q and A

Perceptual quality

z

Quality Adapter
from pretrained P_Q

a

Residual feature → global quality head
r = z + α · sg(‖z‖) · normalize(a)

Q_global

z bypass · α learned, init 0.1

Scalar quality fusion
Q = (1−β) Q_global + β Q_local
β: learned scalar
OUTPUT: PERCEPTUAL QUALITY

P_0

P_0 + projected NSS(image)
Patch scores + NSS weighting
Weighted aggregation → Q_local

Q_local

Text–image alignment

z → image adapter A_G
v: 256-D

Global pair head
[v, u, |v−u|] → A_global

A_global, v, u

U(text) → text adapter A_T
u: 256-D

u

P_0 → independent patch adapter A_P
X_I / X_T matching(u); attention pooling → e
Local pair head [e, u, |e−u|] → A_local

A_local

Sample-wise Alignment gate γ
Inputs: v, u, A_global, A_local
A = γ A_global + (1−γ) A_local
OUTPUT: TEXT–IMAGE ALIGNMENT

TRAINING ONLY a → image-to-text map ↔ five text references → CE (λ = 0.005); score blend = 0

Transfer: compatible backbone, contexts, alignment adapters, quality bank; task heads and layer3 local projection newly initialized.